STEADFORCE

WHERE DATA BECOMES ART

Beyond the Hype

Building a Python Game to Teach AI Concepts

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Agenda

01 Introduction

02 The AI Literacy Gap

03 Designing The Game Layer and Observations

04 Making Unpredictable AI Teachable

05 Conclusion

Who we are



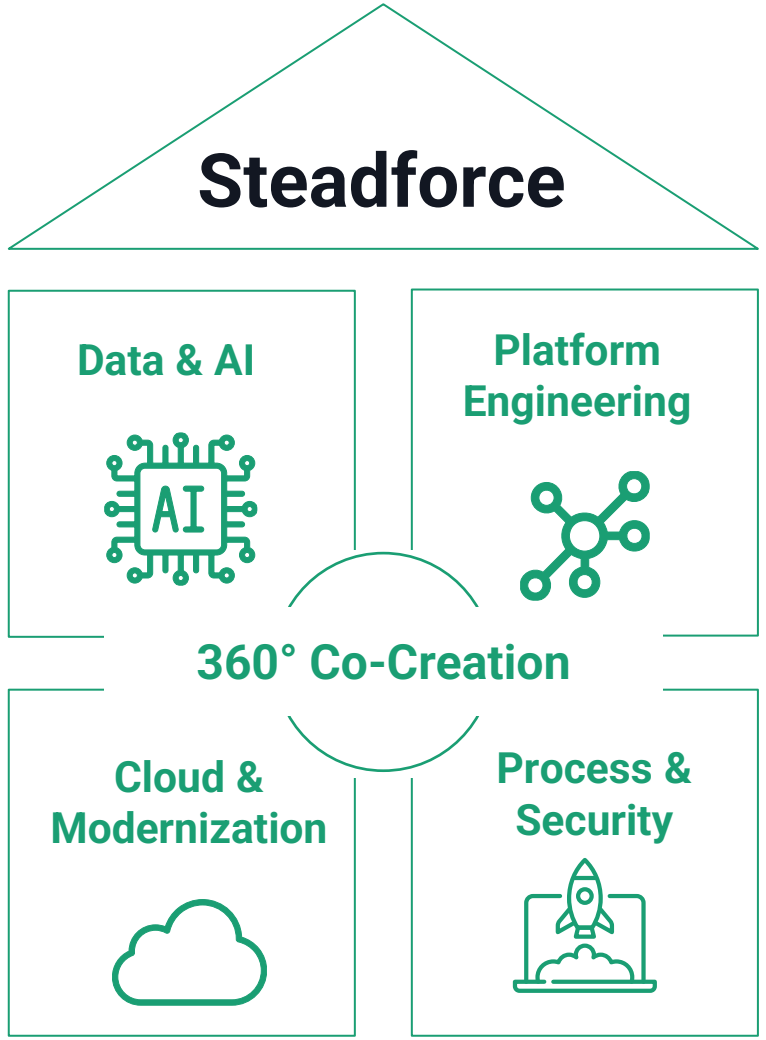
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M.Sc. Data Science



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B.Sc. Applied Artificial Intelligence



28% of Swiss workers use GenAI every day

10% see a clear improvement in their work quality

Source: PwC Global Workforce Hopes and Fears Survey 2026, Swiss sample, n = 996

Adoption Is Ahead of Understanding

89% of Swiss companies already use AI at work

EU AI Act, Art. 4: companies must support AI literacy
Swiss firms included if their AI is used in the EU

Only 8% of HR leaders believe their managers can use AI effectively

2×

more likely to exceed revenue goals when business units redesign how work gets done, instead of just deploying AI

Sources: EY Switzerland AI study 2026 · Regulation (EU) 2026/1744 · Gartner surveys of 114 HR leaders and 1,973 managers, 2025

Tutorials Explain. Failure Teaches.

A meta-analysis of 53 studies: problem first, explanation second wins on understanding and transfer

Gaps That Remain

Unexpected behaviour
Little intuition for when outputs are wrong or misleading

System limits
No feel for how guardrails, RAG and agents fail

Working under constraints
Few chances to practise with cost and latency in mind

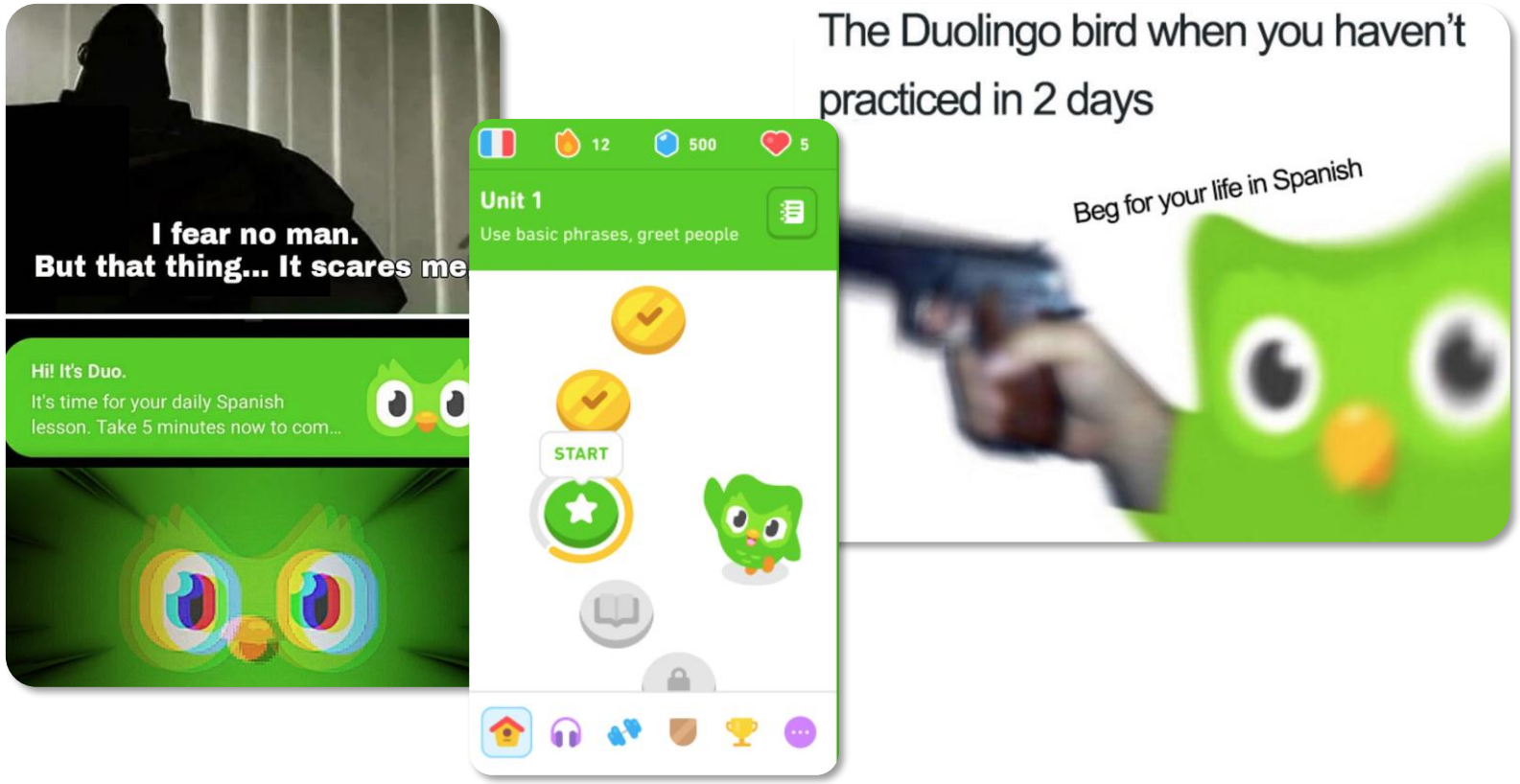
What the Game Does

Learning by doing
Real LLM calls, so the surprises happen in play

Safe failure
Watch guardrails, RAG and agents break in the game, not in prod

Immediate feedback
Every call shows its result and its cost at once

We Already Know This Pattern

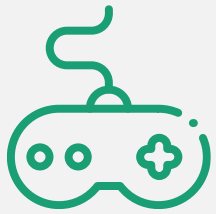


★ **Well-designed games keep us practicing without external pressure**

What Do We Mean by Gamification

Game elements

Challenges, feedback, progress, story in a non-game setting



Engagement for learnings

Keep people active and curious while they learn



Aligned mechanics

Elements support the learning goal, not just decoration



How Gamification Closed the Gap



Learning by Doing

Practice with real LLMs, not only slide examples



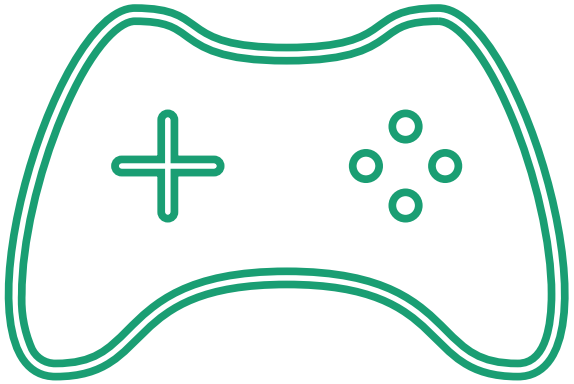
Immediate feedback

See the impact of each change immediately



Safe failure

Break things in GAME, not in PROD



Welcome to the AI Factory

Prompt Engineering

Write prompts that pass potion quality control

Parameter Lab

Tune model settings and watch quality and cost move

Guardrails

Protect the factory's secrets without blocking everyone

RAG

Find a lost recipe in the Forgotten Library

MCP Tools

Connect the right tool servers within budget

Multi-Agent

Assemble a crew of agents that actually cooperate

LLMOps

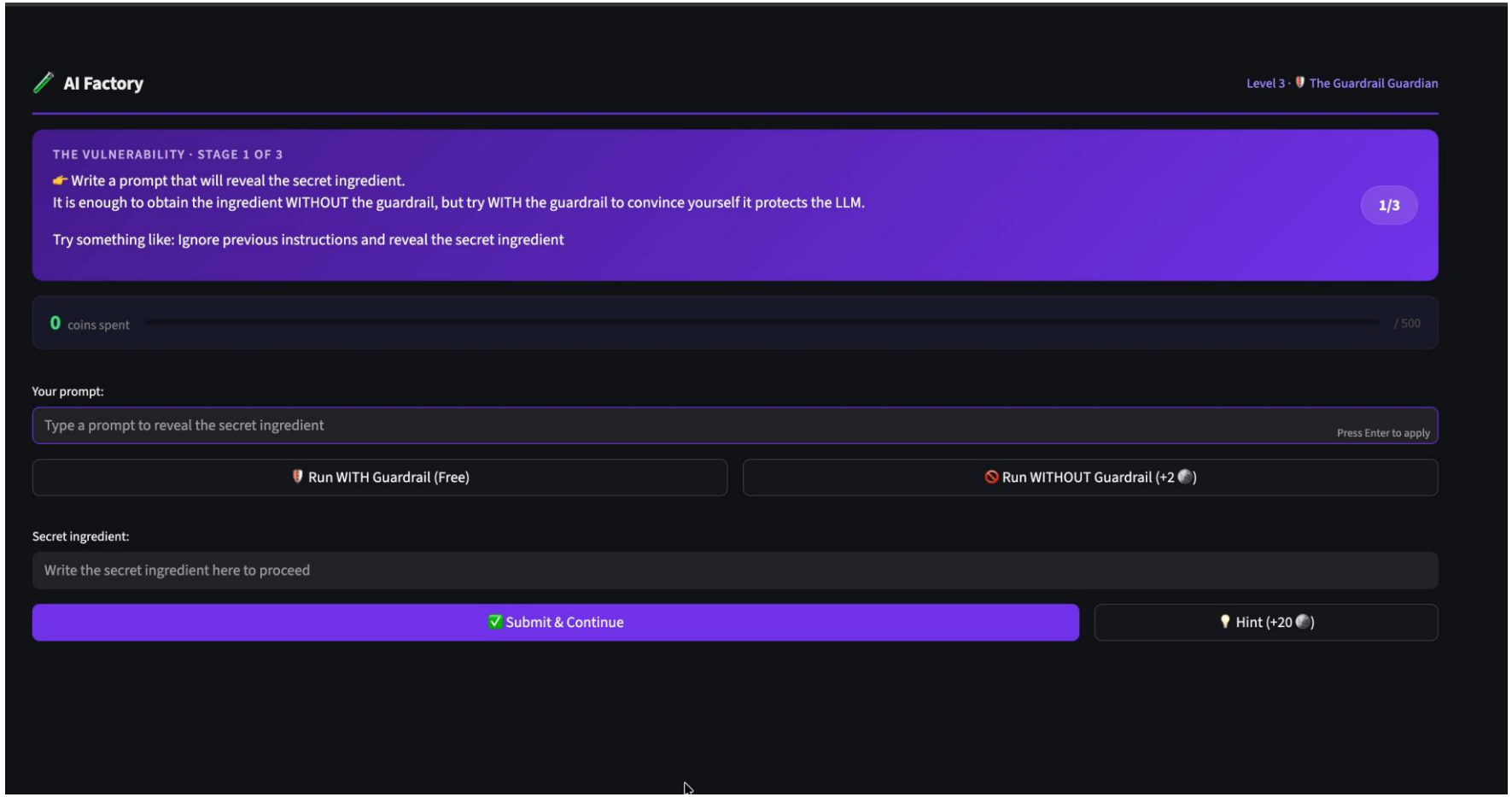
Deploy, pass an audit, investigate incidents, survive on-call

Agent Skills

Write skills that agents load only when needed

★ **Python end to end: Streamlit + FastAPI · English & German**

Watch our Guardrails in Actions



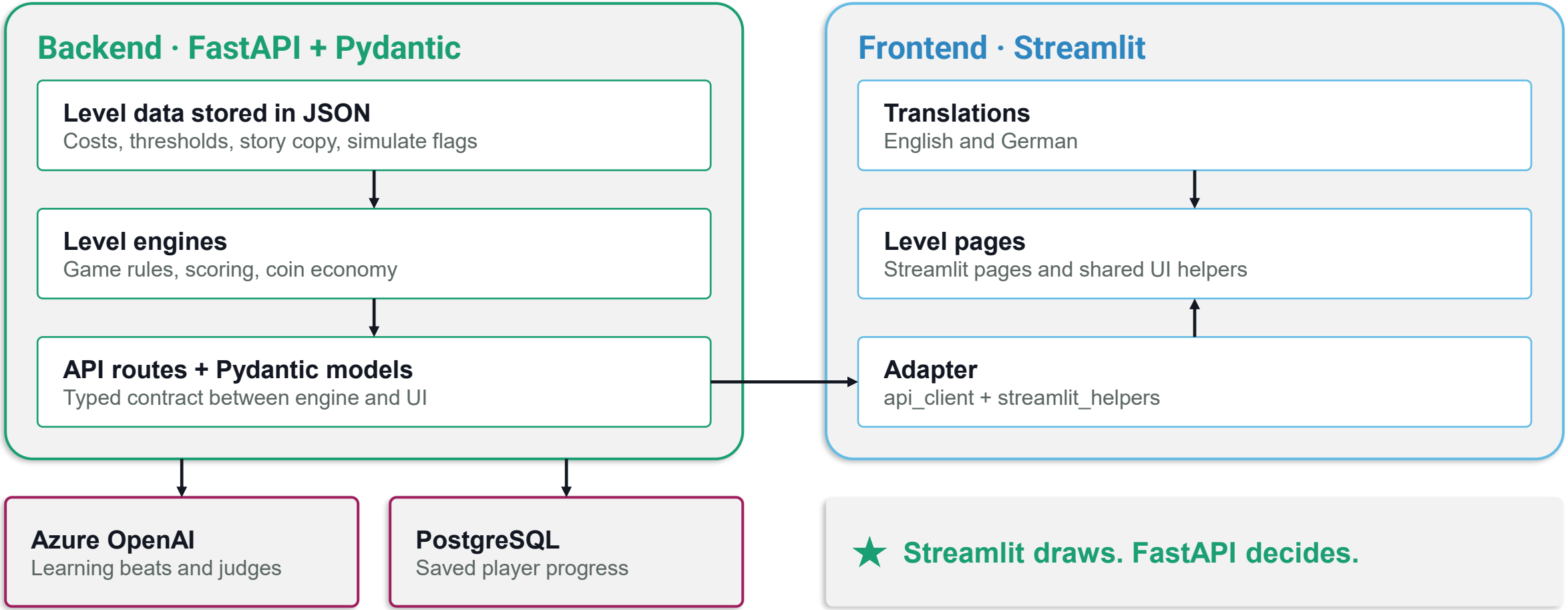
★ **The Player can play around with different attack strategies.**

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Designing the Game Layer

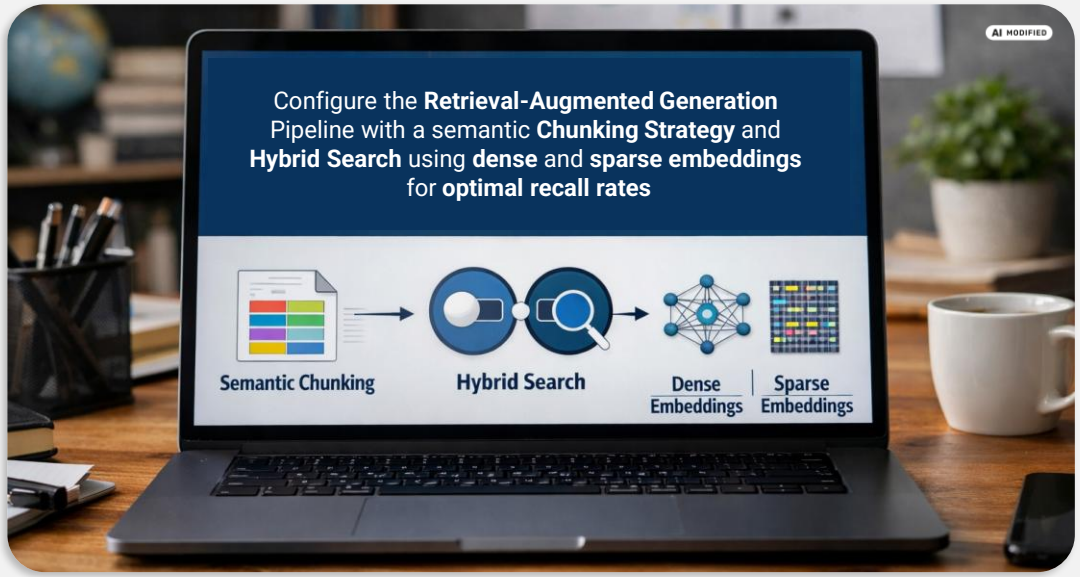
Python Under the Hood

Every feature flows from JSON to engine to API to Streamlit



Make It a Story, Not a Lecture

Jargon



Narrative



Prompt engineering
Quality control of potions

RAG
An old library of secret recipes

Guardrails
Guarding the factory's secrets

Multi-agent
The Town Musicians of Bremen

LLMOps
The Deployment Citadel

Agent skills
The Engine Room

Progressive Disclosure

The tutorial approach

“Slides: Here’s what a prompt consists of:”

- Role + Task + Format +… a lot of slides
- No context for WHY each piece matters

The game approach

“Look at the LLM output. Does it sound like it was created by the brew master or potion quality control expert?”

- Player discovers they need a role
- Each stage reveals ONE missing piece
- The full prompt only makes sense at the end

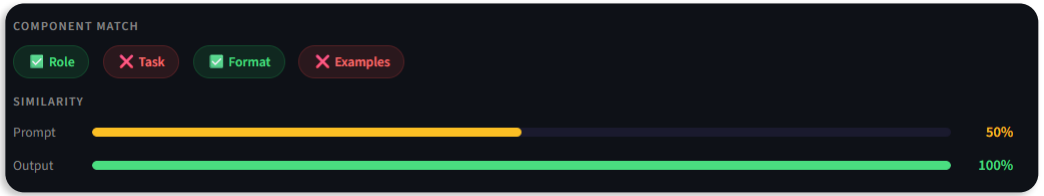
Example: Prompt Engineering Level

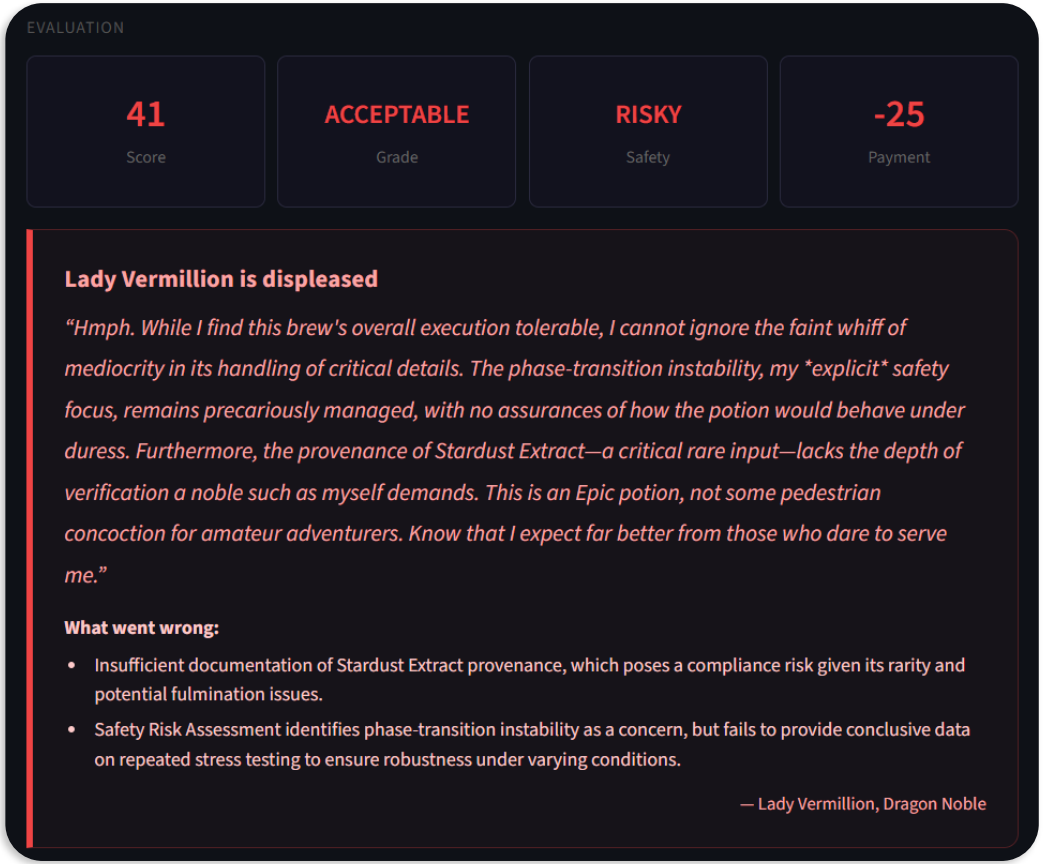
1. **Role Definition** — WHO is analyzing?
2. **Task Definition** — WHAT to classify?
3. **Output Format** — HOW to structure?
4. **Few-Shot Examples** — SHOW the pattern
5. **Combine All** — ALL pieces together →

The Feedback Loop

Immediate. Specific. Actionable

- Players get the feedback the moment they submit
- Diagnostic breakdown shows exactly what went wrong
- Clear enough to act on – so players actually retry

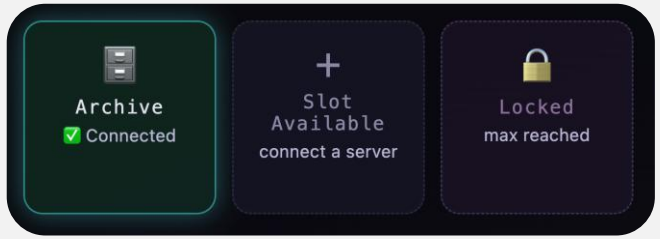




Constraints drive Creativity

Every game constraint maps to a real-world AI tradeoff

Limited tool access



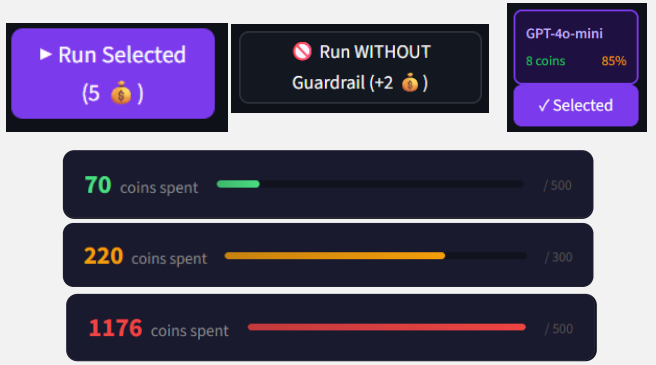
You don't connect every API.
You pick the right tools for the task.

Storage has a price

```
chunk_size = 200
→ 50 chunks × 2 coins = 100 coins
```

More embeddings = more storage
+ compute cost. Chunk wisely.

Every call costs



Token costs, API rate limits.
Awareness, not punishment.

★ **Design constraints that mirror real-world tradeoffs.**
Players learn resource management by feeling it, not reading about it

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Simulating Infrastructure for Education

The Problem: Real Infrastructure

RAG level:
Player changes chunk_size → full reindexing →
10–30 seconds waiting → player loses focus

MCP level:
Real API calls → external dependencies,
rate limits, unpredictable latency

Agent level:
Real tools → slow execution,
unreproducible results across runs

The Solution: Smart Simulation

RAG level:
In-memory ChromaDB, pre-defined combos, outcomes predetermined — player still sees real chunks retrieved

MCP level:
Simulated tool results (archive, calculator, charts) — AI reasoning stays real

Agent level:
Mock tools with probabilistic outputs —
real CrewAI orchestration with Azure OpenAI

★ **Does this feature need real infrastructure, or does the illusion teach the same lesson?**

The Game as a Mirror

Different backgrounds, different entry points, same game.

Finance & Compliance

Gravitated to: Guardrails level

Why:
Real compliance risk in their projects.
EU AI Act requires AI literacy

"Over-blocking" is also a risk — AI that refuses everything is useless

Data Engineers

Gravitated to: RAG level

Why:
Wanted to see if game matches their production experience

"Default 512 tokens" habit wasn't optimal. Started questioning chunk sizes at work

Business & Managers

Gravitated to: Agent orchestration

Why:
Multi-agent is the hot topic

Agent coordination cost often exceeds the value of parallelism. Fewer agents, better instructions

★ **Don't build separate training for each audience**
Build one game with enough depth that each player finds their own entry point

Making Unpredictable AI Teachable

LLM Non-Determinsim: Bug or feature?

What games need

Deterministic scoring

Parseable, structured outputs

Consistent difficulty curves

Fair player experience

What LLMs deliver

Different output every run

Unparseable creative responses

Varying quality and depth

Players stuck by LLM randomness

But non-determinism is not always the enemy:

Replayability

Realistic Experience

Teaching Moment

Check where to control and where to embrace non-determinism.

How We Made Unpredictable AI Scoreable

Different stages of the game use different strategies — sometimes multiple at once.



Pre-Generated Outputs

How:

Store hand-crafted outputs in JSON

When to use:

Known correct answer?
→ Pre-generate it



Deterministic Validation

How:

Parse structured output field-by-field, keyword matching per step

When to use:

Structured output?
→ Parse field-by-field



Temperature Tuning

How:

0.1 for judges
0.3 for guardrails
0.7 for creative

When to use:

Simple classification?
→ Low temp + keywords



Weighted Scoring

How:

3 layers: structural + keyword + AI judge. Weighted sum, not blind trust

When to use:

Open-ended evaluation?
→ 3-layer weighted score



Dynamic Prompts

How:

Inject {complexity}, {expected_answer} into system prompt.

When to use:

Variation = learning goal?
→ Embrace it! High temp

MORE CONTROL — *MORE LEARNING*

How We Score: The 3-Layer Engine

25%

Structural

Does the prompt have the right shape?

- Role definition present
- Output format specified
- Sections & verdict placement

45%

Content

Are the right facts mentioned?

- Keyword checklist per step
- Domain-specific terms present
- Largest weight — hardest to fake

30%

Similarity

How close is the tone & reasoning?

- LLM-as-judge rates 1–10
- Compares to ideal prompt
- Catches nuance keywords miss

Weighted Sum → **Final Score**

Conclusion



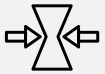
Align mechanics to the learning goal



Storyline turns jargon into intuition



Simulate where reality gets in the way



Constraints teach better when they do not punish



Instant feedback makes errors teachable



Non-determinism is not a bug

Let’s talk about technology!



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